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COMBINED FUNCTION IC DEVICE AND METHOD OF FABRICATING THE SAME

Abstract

An IC device includes first and second gate structures; first and second conductors overlying at least a portion of each of the first and second gate structures; a plurality of gate structures between the first and second gate structures; a conductive segment between the first and second conductors and overlying the plurality of gate structures; first and second pluralities of active areas between the first and second gate structures. The first and second pluralities of active areas are between the first and second conductors; a first portion of the plurality of gate structures, a first portion of the first plurality of active areas, and the second plurality of active areas are included in a functional circuit; and a second portion of the plurality of gate structures, a second portion of the first plurality of active areas, and the conductive segment are included in a decoupling capacitor or an antenna diode.

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Background/Summary

PRIORITY CLAIM [0001] The present application is a continuation of U.S. patent application Ser. No. 18/482,172, filed Oct. 6, 2023, which is a continuation of U.S. patent application Ser. No. 17/830,601, filed Jun. 2, 2022, now U.S. Pat. No. 11,784,646, issued Oct. 10, 2023, which claims priority to China Patent Application No. 202210585532.0, filed May 27, 2022, each of which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] The ongoing trend in miniaturizing integrated circuits (ICs) has resulted in progressively smaller devices which consume less power, yet provide more functionality at higher speeds than earlier technologies. Such miniaturization has been achieved through design and manufacturing innovations tied to increasingly strict specifications. Various electronic design automation (EDA) tools are used to generate, revise, and verify designs for semiconductor devices while ensuring that IC structure design and manufacturing specifications are met.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1 is a diagram of a plan view of an IC layout and corresponding IC device, in accordance with some embodiments.

[0005] FIG. 2A is a schematic diagram of an IC cell and corresponding IC device, in accordance with some embodiments.

[0006] FIG. 2B is a diagram of a plan view of an IC cell and corresponding IC device, in accordance with some embodiments.

[0007] FIG. 3A is a schematic diagram of an IC cell and corresponding IC device, in accordance with some embodiments.

[0008] FIG. 3B is a diagram of a plan view of an IC cell and corresponding IC device, in accordance with some embodiments.

[0009] FIG. 4 is a flowchart of a method of manufacturing an IC device, in accordance with some embodiments.

[0010] FIG. 5 is a flowchart of a method of generating an IC layout diagram, in accordance with some embodiments.

[0011] FIG. 6 is a block diagram of an IC layout diagram generation system, in accordance with

some embodiments.

[0012] FIG. 7 is a block diagram of an IC manufacturing system, and an IC manufacturing flow associated therewith, in accordance with some embodiments.

DETAILED DESCRIPTION

[0013] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components, values, steps, operations, materials, arrangements, or the like, are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. Other components, values, operations, materials, arrangements, or the like, are contemplated. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0014] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0015] In various embodiments, an IC device based on an IC layout diagram includes at least one IC device based on a layout cell in which a first portion includes elements configured as a primary functional circuit, e.g., a level shifter, and a second portion includes elements configured as a secondary functional circuit, e.g., one of a decoupling capacitor or an antenna diode. Compared to approaches in which cells configured as decoupling capacitors and antenna diodes are separate from those configured as primary functional circuits, overall IC device space is reduced.

[0016] As discussed below, FIGS. 1, 2B, and 3B depict plan views in some embodiments. Each of FIGS. 1, 2B, and 3B is a device/layout diagram in which the reference designators represent both IC device features and the IC layout features used to at least partially define the corresponding IC device features in a manufacturing process, e.g., a method **400** discussed below with respect to FIG. 4 and/or an IC manufacturing flow associated with an IC manufacturing system **700** discussed below with respect to FIG. 7. In some embodiments, one or more of FIG. 1, 2B, or 3B is some or all of an IC layout diagram generated by executing some or all of the operations of a method **500** discussed below with respect to FIG. 5. Accordingly, each of FIGS. 1, 2B, and 3B represents a plan view of both an IC layout diagram and a corresponding IC device.

[0017] Each of the figures herein, e.g., FIGS. 1, 2B, and 3B, is simplified for the purpose of illustration. The figures are views of IC structures and devices with various features included and excluded to facilitate the discussion below. In various embodiments, an IC structure, device, and/or layout diagram includes one or more features corresponding to power distribution structures, metal interconnects, contacts, vias, gate structures, source/drain (S/D) structures, or other transistor elements, isolation structures, or the like, in addition to the features depicted in FIGS. 1, 2B, and 3B.

[0018] FIG. 1 is a diagram of a plan view of an IC layout **100** and a corresponding IC device **100**, in accordance with some embodiments. In addition to the top-level plan view of IC layout/device **100**, FIG. 1 includes X and Y directions.

[0019] IC layout/device **100** includes a region **100F** and cells/devices **110**, **120**, and **130**. Region **100F** includes an arrangement of cells and corresponding devices (not shown individually) having

various sizes and dimensions in accordance with being functional circuits configured to perform predetermined functions, e.g., logic, computation, data and signal processing, data storage, or the like. In some embodiments, a cell is referred to as a layout or layout component. The cells/devices of region **100F** have rectangular shapes including widths in the X direction and heights in the Y direction (not shown). In some embodiments, region **100F** includes multiple power domains corresponding to the various functions of the cells/devices, e.g., one or more domains having a memory power supply voltage level greater than a power supply voltage level of one or more other domains including signal processing devices.

[0020] Each of cells/devices **110**, **120**, and **130** is an IC cell and corresponding IC device having a rectangular shape and including features configured as both a functional circuit, also referred to as a first functional circuit, and one of a decoupling capacitor or an antenna diode, also referred to as a second functional circuit. In some embodiments, the first functional circuit of one or more of cells/devices **110**, **120**, or **130** is a level shifter: a low-to-high level shifter configured to convert signals from a first to second power domain in which the second power domain has a power supply voltage level greater than that of the first power domain; or a high-to-low level shifter configured to convert signals from a first to second power domain in which the first power domain has a power supply voltage level greater than that of the second power domain.

[0021] In some embodiments, one or more of cells/devices **110**, **120**, or **130** is a IC cell/device **200** including features configured as both the first functional circuit and a decoupling capacitor discussed below with respect to FIGS. 2A and 2B. In some embodiments, one or more of cells/devices **110**, **120**, or **130** is a IC cell/device **300** including features configured as both the first functional circuit and an antenna diode discussed below with respect to FIGS. 3A and 3B.

[0022] The number of cells/devices **110**, **120**, and **130** depicted in FIG. 1 is a non-limiting example provided for the purpose of illustration. In various embodiments, IC layout/device **100** does not include one or more of cells/devices **110**, **120**, or **130**, or includes one or more cells/devices (not shown) in addition to cells/devices **110**, **120**, and **130**.

[0023] The positioning and relative sizes of cells/devices **110**, **120**, and **130** within IC layout/device **100** depicted in FIG. 1 is a non-limiting example provided for the purpose of illustration. In various embodiments, IC layout/device **100** includes cells/devices, e.g., cells/devices **110**, **120**, and **130**, otherwise positioned and/or sized within IC layout/device **100**, e.g., positioned along an edge of IC layout/device **100**.

[0024] FIG. 2A is a schematic diagram of IC cell/device **200**, in accordance with some embodiments, and FIG. 3A is a schematic diagram of IC cell/device **300**, in accordance with some embodiments.

[0025] Each of IC cells/devices **200** and **300** includes a first portion **200P1** and a corresponding second portion **200P2** or **300P2**. First portion **200P1** includes p-type metal oxide semiconductor (PMOS) and n-type metal oxide semiconductor (NMOS) transistors (not labeled) configured as a level shifter, also referred to as level shifter **200P1** in some embodiments.

[0026] A first subset of the PMOS and NMOS transistors of level shifter **200P1** is configured to operate in a first power domain including a power supply voltage VDDL and a power supply reference voltage VSS. The first subset includes an inverter configured to receive an input signal IN and generate a signal Ib, and an inverter configured to receive signal Ib and generate a signal Ibb.

[0027] A second subset of the PMOS and NMOS transistors of level shifter **200P1** is configured to operate in a second power domain including a power supply voltage VDD and a power supply reference voltage VSS. Power supply voltage VDD has a power-on voltage level greater than a power-on voltage level of power supply voltage VDDL. The second subset includes an inverter configured to receive an input signal NSLP and generate a signal slp, and an inverter configured to receive signal slp and generate a signal nslp.

[0028] The second subset also includes a level-shifting arrangement of PMOS and NMOS

transistors configured to receive complementary signals Ib and Ibb and generate a signal zi on an internal node (not labeled). The level-shifting arrangement is selectively coupled to power supply reference voltage VSS through an NMOS transistor configured to receive signal nslp, and the internal node is selectively coupled to power supply voltage VDD through a PMOS transistor configured to receive signal nslp. A pair of inverters is configured to receive signal zi and generate an output signal Z.

[0029] Level shifter **200P1** is thereby configured to, in operation, receive and buffer signal IN in the first power domain, and buffer and generate signal Z in the second power domain based on signal IN. In some embodiments, level shifter **200P1** is configured to receive and buffer signal IN and buffer and generate signal Z as clock signals.

[0030] Level shifter **200P1** is thereby configured further to, in operation, receive and buffer signal NSLP in the second power domain, and responsive to signal NSLP, enable and disable the signal buffering and generation in the second power domain. In some embodiments, level shifter **200P1** is configured to receive signal NSLP as a sleep mode or low-power enable signal.

[0031] Second portion **200P2**, also referred to as a decoupling capacitor **200P2** (or decap) in some embodiments, includes an NMOS transistor T1 including a gate configured to receive power supply voltage VDD, and each of source, drain, and substrate terminals configured to receive power supply reference voltage VSS. Transistor T1 is thereby configured as a capacitive device including a gate capacitance coupled between power supply voltage VDD received at the gate and power supply reference voltage VSS received at each of the other terminals. In operation, the gate capacitance acts to decouple noise from adjacent functional circuits, e.g., level shifter **200P1**, configured to receive power supply voltage VDD.

[0032] In some embodiments, transistor T1 includes a plurality of NMOS transistors configured in parallel such that a total capacitance of transistor T1 is equal to a sum of the gate capacitances of each transistor of the plurality of NMOS transistors. As a number of parallel transistors of transistor T1 increases, the total capacitance of transistor T1 thereby also increases such that the capacity to decouple power supply voltage VDD noise also increases.

[0033] In some embodiments, the number of parallel transistors of transistor T1 ranges from five to 50. In some embodiments, the number of parallel transistors of transistor T1 ranges from 15 to 30.

[0034] Second portion **300P2**, also referred to as antenna diode **300P2** in some embodiments, includes an NMOS transistor T2 including a gate and each of source and drain terminals configured to receive input signal IN, and a substrate terminal configured to receive power supply reference voltage VSS. Transistor T2 is thereby configured as a diode device including a forward conduction path from signal IN received at the gate and source and drain terminals to power supply reference voltage VSS received at the substrate terminal. In operation, the forward conduction path acts to divert potentially damaging current levels at inputs to adjacent functional circuits, e.g., level shifter **200P1**, configured to receive input signal IN.

[0035] In some embodiments, transistor T2 includes a plurality of NMOS transistors configured in parallel such that a total current diverting capacity of transistor T2 is equal to a sum of the current diverting capacities of each transistor of the plurality of NMOS transistors. As a number of parallel transistors of transistor T2 increases, the total capacity of transistor T2 to divert potentially damaging current levels from input signal IN also increases.

[0036] In some embodiments, the number of parallel transistors of transistor T2 ranges from five to 50. In some embodiments, the number of parallel transistors of transistor T2 ranges from 15 to 30.

[0037] FIG. 2B is a diagram of a plan view of IC cell/device **200**, in accordance with some embodiments, and FIG. 3B is a diagram of a plan view of IC cell/device **300**, in accordance with some embodiments. The embodiments depicted in FIGS. 2B and 3B are non-limiting examples of respective IC cells/devices **200** and **300** provided for the purpose of illustration. Other configurations of IC cell/device **200** and/or **300**, e.g., including a functional circuit other than a level shifter or including fewer or greater numbers of parallel transistors, are within the scope of

the present disclosure.

[0038] Each of IC cells/devices **200** and **300** includes n-well regions/structures **N1-N3** positioned in a semiconductor wafer (not labeled), pluralities of active regions/areas **AA1** to **AA4** extending in the X direction in the semiconductor wafer, instances of each of gate regions/structures **G1-G4** extending in the Y direction, power rails **VSS** and **VDD** extending in the X direction, and instances of via regions/structures **V0** and metal regions/segments **M1** (single instances labeled for clarity) extending in the Y direction. An instance of metal region/segment **M1** is configured to receive power supply voltage **VDDL**, an instance of metal region/segment **M1** is configured to receive input signal **IN**, and an instance of metal region/segment **M1** is configured to carry output signal **Z**. Plurality of active regions/areas **AA1** includes active regions/areas **A1-A5** aligned in the X direction, and pluralities of active regions/areas **AA2-AA4** include active regions/areas that are not labeled for the purpose of clarity.

[0039] IC cell/device **200** also includes instances of via regions/structures **VG** and a metal region/segment **M02** extending in the X direction, and IC cell/device **300** also includes instances of via regions/structures **VD** and a metal region/segment **M03** extending in the X direction.

[0040] As discussed below, a first portion of the features of each of IC cells/devices **200** and **300** corresponds to first portion **200P1**, a second portion of the features of IC cell/device **200** corresponds to second portion **200P2**, and a second portion of the features of IC cell/device **300** corresponds to second portion **300P2**.

[0041] A well/region, e.g., an n-well **N1-N3**, is a region in an IC layout diagram corresponding to an IC device, e.g., IC cell/device **200** or **300**, included in a manufacturing process as part of defining a well, i.e., a continuous portion of a semiconductor wafer, e.g., a silicon (Si) wafer or an epitaxial Si layer, suitable for forming one or more IC devices, e.g., IC devices **100-300**. In various embodiments, a well is a p-well based on the semiconductor portion including one or more acceptor dopants, e.g., boron (B) or aluminum (Al), or an n-well based on the semiconductor portion including one or more donor dopants, e.g., phosphorous (P) or arsenic (As). In the embodiments depicted herein, each of IC cells/devices **200** and **300** includes n-wells **N1-N3**. In various embodiments, an IC cell/device **200** and/or **300** includes one or more p-wells in addition to or instead of one or more of n-wells **N1-N3**, includes one or more n-wells in addition to n-wells **N1-N3**, and/or does not include one or more of n-wells **N1-N3**.

[0042] N-well **N2** corresponds to the first power domain and each of n-wells **N1** and **N3** corresponds to the second power domain. N-well **N2** is separated from n-wells **N1** and **N3** in the X direction by one or more distances in accordance with minimum spacing rules of the manufacturing process corresponding to latch-up prevention based on power supply voltage levels of the first and second power domains.

[0043] An active area, e.g., an active area of pluralities of active regions/areas **AA1-AA4**, is a region in the IC layout diagram included in the manufacturing process as part of defining an active area, also referred to as an oxide diffusion or definition (OD), in the semiconductor wafer, either directly or in one of n-wells **N1-N3**, in which one or more IC device features, e.g., a source/drain structure (not shown), is formed. In some embodiments, an active area is an n-type or p-type active area of a planar transistor, a fin field-effect transistor (FinFET), or a gate-all-around (GAA) transistor. In various embodiments, an active area (structure) includes one or more of a semiconductor material, e.g., silicon (Si), silicon-germanium (SiGe), silicon-carbide (SiC), or the like, an acceptor or donor dopant material, or another suitable material.

[0044] In some embodiments, an active area is a region in an IC layout diagram included in the manufacturing process as part of defining a nano-sheet structure, e.g., a continuous volume of one or more layers of one or more semiconductor materials having either n-type or p-type doping. In various embodiments, individual nano-sheet layers include a single monolayer or multiple monolayers of a given semiconductor material.

[0045] Each of pluralities of active regions/areas **AA1** and **AA4** includes n-type active

regions/areas adjacent to an instance of power rail VSS, and each of pluralities of active regions/areas AA1 and AA4 includes p-type active regions/areas positioned in corresponding n-wells N1-N3 adjacent to power rail VDD.

[0046] A gate region/structure, e.g., a gate regions/structures G1-G4, is a region in the IC layout diagram included in the manufacturing process as part of defining a gate structure. A gate structure is a volume including one or more conductive segments, e.g., a gate electrode, including one or more conductive materials, e.g., polysilicon, copper (Cu), aluminum (Al), tungsten (W), cobalt (Co), ruthenium (Ru), or one or more other metals or other suitable materials, substantially surrounded by one or more insulating materials, the one or more conductive segments thereby being configured to control a voltage provided to an adjacent gate dielectric layer.

[0047] A dielectric layer, e.g., a gate dielectric layer, is a volume including one or more insulating materials, e.g., silicon dioxide, silicon nitride (Si.sub.3N.sub.4), and/or one or more other suitable material such as a low-k material having a k value less than 3.8 or a high-k material having a k value greater than 3.8 or 7.0 such as aluminum oxide (Al.sub.2O.sub.3), hafnium oxide (HfO.sub.2), tantalum pentoxide (Ta.sub.2O.sub.5), or titanium oxide (TiO.sub.2), suitable for providing a high electrical resistance between IC structure elements, i.e., a resistance level above a predetermined threshold corresponding to one or more tolerance levels of a resistance-based effect on circuit performance.

[0048] Gate regions/structures G1-G4 have configurations along the Y direction in accordance with cut-gate regions CP1-CP5 (borders not depicted in FIGS. 2B and 3B for the purpose of clarity), also referred to as cut-poly regions CP1-CP5 in some embodiments. A cut-gate region is a region in the IC layout diagram included in the manufacturing process as part of defining a discontinuity in the gate electrode of a given gate structure, thereby electrically isolating the corresponding adjacent portions of the gate electrode from each other.

[0049] Each of cut-gate regions CP1 and CP5 extends continuously across the widths of each of IC cells/devices 200 and 300 in the X direction such that each of gate regions/structures G1-G4 has endpoints in the Y direction overlapping/underlying the instances of power rails VSS.

[0050] Each of cut-gate regions CP2-CP4 includes multiple portions extending across the widths of each of IC cells/devices 200 and 300 in the X direction such that the instances of gate regions/structures G1-G4 have from zero through three discontinuities in the Y direction. Each instance of gate region/structure G1 has zero discontinuities corresponding to the portions of cut-gate regions CP2-CP4 and includes a single gate electrode extending in the Y direction between the instances of power rail VSS. Each instance of gate region/structure G2 has one discontinuity corresponding to a portion of one of cut-gate regions CP2-CP4 and includes a total of two gate electrode portions extending in the Y direction between the instances of power rail VSS. Each instance of gate region/structure G3 has two discontinuities corresponding to portions of two of cut-gate regions CP2-CP4 and includes a total of three gate electrode portions extending in the Y direction between the instances of power rail VSS. Each instance of gate region/structure G4 has three discontinuities corresponding to portions of three of cut-gate regions CP2-CP4 and includes a total of four gate electrode portions extending in the Y direction between the instances of power rail VSS.

[0051] The number and positions of cut-gate regions CP1-CP5 are non-limiting examples provided for the purpose of illustration. In various embodiments, one or both of IC cells/devices 200 or 300 includes gate regions/structures having portions based on fewer or greater than five cut-gate regions and/or based on cut-gate regions at locations other than those depicted in FIGS. 2B and 3B.

[0052] A metal region/segment, e.g., a metal region/segment M02, M03, or M1 or power rail VSS or VDD, is a region in the IC layout diagram included in the manufacturing process as part of defining a metal segment structure including one or more conductive materials in a given metal layer of the manufacturing process. In various embodiments, each of power rails VSS and VDD corresponds to a first metal layer, metal regions/segments M02 and M03 correspond to the first

metal layer, and/or metal regions/segments **M1** corresponds to a second metal layer of the manufacturing process.

[0053] A via region/structure, e.g., a via region/structure **VD**, **VG**, or **V0**, is a region in the IC layout diagram included in the manufacturing process as part of defining a via structure including one or more conductive materials configured to provide an electrical connection between an overlying conductive structure, e.g., a power rail or other metal segment, and an underlying conductive structure. The underlying structure corresponds to a S/D structure in the case of a via region/structure **VD**, a gate electrode in the case of a via region/structure **VG**, and a first metal layer region/segment, e.g., metal region/segment **M02** or **M03**, in the case of a via region/structure **V0**. The depictions of via regions/structures **VD** in FIG. 3B are considered to include the underlying S/D structures which are not separately depicted for the purpose of clarity.

[0054] First portion **200P1** includes n-wells **N1-N3**, active regions/areas **A1** and **A5** of plurality of active regions/areas **AA1**, each of pluralities of active regions/areas **AA2-AA4**, the portions of gate regions/structures **G1-G4** outside the boundary of second portion **200P2** or **300P2**, the instances of metal regions/segments **M1** and via regions/structures **V0**, and additional features that are not depicted for the purpose of clarity. The various features are configured as pluralities of PMOS and NMOS transistors (not labeled) arranged in accordance with the schematic diagrams of FIGS. 2A and 3A discussed above such that first portion **200P1** includes a functional circuit, e.g., a level shifter.

[0055] The locations and configurations of the instances of metal regions/segments **M1** of first portion **200P1** corresponding to power supply voltage **VDDL**, input signal **IN**, and output signal **Z** depicted in FIGS. 2B and 3B are non-limiting examples provided for the purpose of illustration. In various embodiments, first portion **200P1** includes instances of metal regions/segments **M1** corresponding to one or more of power supply voltage **VDDL**, input signal **IN**, or output signal **Z** otherwise configured in accordance with the schematic diagrams of FIGS. 2A and 3A.

[0056] Horizontal boundaries of first portion **200P1** correspond to endmost instances of gate regions/structures **G1-G4**, each of the endmost instances being an instance of gate region/structure **G1** extending continuously in the Y direction between the instances of power rail **VSS**. Each of pluralities of active regions/areas **AA1-AA4** extends in the X direction between the endmost instances of gate regions/structures **G1-G4**, the endmost instances thereby defining a width of each of first portion **200P1** and IC cells/devices **200** and **300** in the X direction (not labeled).

[0057] Based on the minimum spacing rules and widths of n-wells **N1-N3**, first portion **200P1** and IC cells/devices **200** and **300** have the width sufficiently large such that portions of the features of each of IC cells/devices **200** and **300** are unused in the level shifter of FIGS. 2A and 3A, and at least some of the unused features correspond to either second portion **200P2** of IC cell/device **200** or second portion **300P2** of IC cell/device **300**.

[0058] Second portion **200P2** of IC cell/device **200** includes active regions/areas **A2-A4** of plurality of active regions/areas **AA1**, the portions of gate regions/structures **G1-G4** inside the boundary of second portion **200P2**, metal region/segment **M02**, instances of via regions/structures **VG**, configured in accordance with the schematic diagram of FIG. 2A as discussed below.

[0059] Second portion **200P2** includes an electrical connection (not shown for the purpose of clarity) between metal region/segment **M02** and a metal region/segment configured to carry power supply voltage **VDD**, e.g., power rail **VDD**. Each instance of via region/structure **VG** is positioned between and configured to electrically connect metal region/segment **M02** and an underlying gate structure portion of gate regions/structures **G1-G4**. Based on cut-gate region **CP2**, each of the corresponding gate structure portions is aligned in the Y direction with, and electrically isolated from, a gate structure portion of the portions of gate regions/structures **G1-G4** included in first portion **200P1**.

[0060] Second portion **200P2** includes one or more electrical connections (not shown for the purpose of clarity) from substrate portions and S/D structures (not shown for the purpose of clarity)

of active areas A2-A4 adjacent to the corresponding gate structure portions to one or more metal regions/segments configured to carry power supply reference voltage VSS, e.g., an instance of power rail VSS.

[0061] The features of second portion **200P2** are thereby arranged as transistor T1 including a plurality of NMOS transistors configured in parallel as a decoupling capacitor. In the embodiment depicted in FIG. 2B, transistor T1 includes a total of seventeen parallel transistors. In some embodiments, transistor T1 includes fewer or greater than seventeen parallel transistors, as discussed above with respect to FIG. 2A.

[0062] Second portion **300P2** of IC cell/device **300** includes active regions/areas A2-A4 of plurality of active regions/areas AA1, the portions of gate regions/structures G1-G4 inside the boundary of second portion **300P2**, metal region/segment M03, instances of via regions/structures VD, configured in accordance with the schematic diagram of FIG. 3A as discussed below.

[0063] Second portion **300P2** includes the instance of metal region/segment configured to carry input signal IN electrically connected to metal region/segment M03 through an instance of via region/structure V0. Each instance of via region/structure VD is positioned between and configured to electrically connect metal region/segment M03 and an underlying S/D structure adjacent to a portion of gate regions/structures G1-G4. Based on cut-gate region CP2, each of the corresponding gate structure portions is aligned in the Y direction with, and electrically isolated from, a gate structure portion of the portions of gate regions/structures G1-G4 included in first portion **200P1**. In some embodiments, second portion **300P2** includes instances of via region/structure VG (not shown) positioned between and configured to electrically connect metal region/segment M03 (or another parallel and similarly configured metal region/segment-not shown) and the corresponding gate structure portions.

[0064] Second portion **300P2** includes one or more electrical connections (not shown for the purpose of clarity) from substrate portions adjacent to the corresponding S/D structures to one or more metal regions/segments configured to carry power supply reference voltage VSS, e.g., an instance of power rail VSS.

[0065] The features of second portion **300P2** are thereby arranged as transistor T2 including a plurality of NMOS transistors configured in parallel as an antenna diode. In the embodiment depicted in FIG. 3B, transistor T2 includes a total of fifteen parallel transistors. In some embodiments, transistor T1 includes fewer or greater than fifteen parallel transistors, as discussed above with respect to FIG. 3A.

[0066] Each of IC cell/device **200** and **300** thereby includes first portion **200P1** including features configured as a primary functional circuit, e.g., a level shifter, and second portion **200P2** or **300P2** including features configured as a secondary functional circuit, e.g., decoupling capacitor T1 or antenna diode T2. Compared to approaches in which cells configured as decoupling capacitors and antenna diodes are separate from those configured as primary functional circuits, overall IC device space is thereby reduced.

[0067] In some embodiments in which a cell/device includes a decoupling capacitor, e.g., IC cell/device **200** including decoupling capacitor T1, device space is reduced by 23% while an increase in power supply voltage VDD leakage is limited to 0.5%, a total power increase is limited to 2.5%, a delay increase is limited to 0.6%, and other device performance metrics are not significantly affected, compared to other approaches. In some embodiments in which a cell/device includes an antenna diode, e.g., IC cell/device **300** including antenna diode T2, device space is reduced by 25% while other device performance metrics are not significantly affected, compared to other approaches.

[0068] FIG. 4 is a flowchart of a method **400** of manufacturing an IC device, in accordance with some embodiments. Method **400** is operable to form the IC device **100** discussed above with respect to FIG. 1, and/or IC devices **200** and/or **300** discussed above with respect to FIGS. 2A-3B.

[0069] In some embodiments, the operations of method **400** are performed in the order depicted in

FIG. 4. In some embodiments, the operations of method **400** are performed in an order other than the order depicted in FIG. 4. In some embodiments, one or more additional operations are performed before, during, and/or after the operations of method **400**. In some embodiments, performing some or all of the operations of method **400** includes performing one or more operations as discussed below with respect to IC manufacturing system **700** and FIG. 7.

[0070] At operation **410**, in some embodiments, first through fourth pluralities of active areas extending in a first direction are formed. In some embodiments, forming the first through fourth pluralities of active areas includes forming pluralities of regions/active areas AA1-AA4 discussed above with respect to FIGS. 1-3B.

[0071] In some embodiments, forming the first through fourth pluralities of active areas includes forming the second and third pluralities of active areas in one or more n-wells, e.g., n-wells N1-N3 discussed above with respect to FIGS. 1-3B.

[0072] In some embodiments, forming the first through fourth pluralities of active areas includes performing one or more implantation processes in areas of a semiconductor wafer corresponding to the first through fourth pluralities of active areas, whereby predetermined doping concentrations and types are achieved for one or more given dopants as discussed above with respect to FIGS. 2B and 3B.

[0073] In some embodiments, forming the first through fourth pluralities of active areas includes forming a plurality of S/D structures in and/or on some or all of the active areas of the first through fourth pluralities of active areas, e.g., by performing one or more implantation processes and/or one or more deposition processes.

[0074] At operation **420**, in some embodiments, a plurality of gate structures extending in a second direction is constructed whereby the active areas of the first through fourth pluralities of active areas are aligned between endmost gate structures of the plurality of gate structures. In some embodiments, constructing the plurality of gate structures includes constructing gate structures G1-G4 discussed above with respect to FIGS. 1-3B.

[0075] In some embodiments, constructing the plurality of gate structures includes performing a plurality of manufacturing operations, e.g., one or more of a lithography, diffusion, deposition, etching, planarizing, or other operation suitable for constructing the plurality of gate structures as discussed above with respect to FIGS. 2B and 3B.

[0076] At operation **430**, in some embodiments, first through third power rails and a conductive segment extending in the first direction are formed by overlying the first endpoint of each of the first and second endmost gate structures with the first power rail and overlying the second endpoint of each of the first and second endmost gate structures with the second power rail.

[0077] In some embodiments, forming the first through third power rails includes forming instances of power rails VSS and power rail VDD discussed above with respect to FIGS. 2B and 3B. In some embodiments, forming the conductive segment extending in the first direction includes forming metal segment M02 discussed above with respect to FIG. 2B or metal segment M03 discussed above with respect to FIG. 3B.

[0078] In some embodiments, forming a metal segment, e.g., the first through third power rails or conductive segment, includes performing a plurality of manufacturing operations including depositing and patterning one or more photoresist layers, performing one or more etching processes, and performing one or more deposition processes whereby one or more conductive materials are configured to form a continuous, low resistance structure.

[0079] The first through third power rails, a first portion of the plurality of gate structures, and a first portion of the first through fourth pluralities of active areas are configured as a functional circuit, and the conductive segment, a second portion of the plurality of gate structures, and a second portion of the first through fourth pluralities of active areas are configured as one of a decoupling capacitor or an antenna diode.

[0080] In some embodiments, the first through third power rails, first portion of the plurality of

gate structures, and first portion of the first through fourth pluralities of active areas are configured as the functional circuit including a level shifter, e.g., first portion **200P1** discussed above with respect to FIGS. **2A-3B**.

[0081] In some embodiments, the conductive segment and the second portions of the plurality of gate structures and first through fourth pluralities of active areas are configured as the decoupling capacitor including decoupling capacitor **200P2** discussed above with respect to FIG. **2B**.

[0082] In some embodiments, forming the first through third power rails and the conductive segment includes configuring the decoupling capacitor by constructing vias between gate structures of the second portion of the plurality of gate structures and the conductive segment and forming an electrical connection between the conductive segment and the third power rail, e.g., constructing via structures **VG** and forming an electrical connection to metal segment **M02** as discussed above with respect to FIG. **2B**.

[0083] In some embodiments, forming the first through third power rails and the conductive segment includes configuring the conductive segment and the second portions of the plurality of gate structures and first through fourth pluralities of active areas as the antenna diode by configuring antenna diode **300P2** discussed above with respect to FIG. **3B**.

[0084] In some embodiments, configuring the antenna diode includes constructing vias between S/D structures in active areas of the second portion of the first plurality of active areas and the conductive segment and forming an electrical connection between the conductive segment and a signal node of the level shifter, e.g., constructing via structures **VD** and forming an electrical connection to metal segment **M03** as discussed above with respect to FIG. **3B**.

[0085] In some embodiments, an active area of the second portion of the first plurality of active areas is aligned between active areas of the first portion of the first plurality of active areas, e.g., active areas **A2-A4** aligned between active areas **A1** and **A5** as discussed above with respect to FIGS. **2B** and **3B**.

[0086] In some embodiments, the plurality of gate structures includes discontinuities between gate structures of the first portion of the plurality of gate structures and gate structures of the second portion of the plurality of gate structures, e.g., discontinuities in gate structures **G1-G4** in accordance with cut-gate regions **CP1-CP5** as discussed above with respect to FIGS. **2B** and **3B**.

[0087] In some embodiments, the functional circuit and the one of the decoupling capacitor or the antenna diode are configured as part of building a plurality of IC devices, e.g., transistors, logic gates, memory cells, interconnect structures, and/or other suitable devices, by performing a plurality of manufacturing operations, e.g., one or more of a lithography, diffusion, deposition, etching, planarizing, or other operation suitable for building the plurality of IC devices in the semiconductor wafer.

[0088] By performing some or all of the operations of method **400**, an IC device is manufactured in which a first portion includes features configured as a primary functional circuit, e.g., a level shifter, and a second portion includes features configured as a decoupling capacitor or antenna diode, thereby obtaining the benefits discussed above with respect to IC devices **100-300**.

[0089] FIG. **5** is a flowchart of a method **500** of generating an IC layout diagram, e.g., an IC layout **100** discussed above with respect to FIG. **1** and/or a IC cell **200** and/or **300** discussed above with respect to FIGS. **2A-3B**, in accordance with some embodiments.

[0090] In some embodiments, generating the IC layout diagram includes generating the IC layout diagram corresponding to an IC device, e.g., an IC device **100-300** discussed above with respect to FIGS. **1-3B**, manufactured based on the generated IC layout diagram.

[0091] In some embodiments, some or all of method **500** is executed by a processor of a computer, e.g., a processor **602** of an IC layout diagram generation system **600**, discussed below with respect to FIG. **6**.

[0092] Some or all of the operations of method **500** are capable of being performed as part of a design procedure performed in a design house, e.g., a design house **720** discussed below with

respect to FIG. 7.

[0093] In some embodiments, the operations of method **500** are performed in the order depicted in FIG. 5. In some embodiments, the operations of method **500** are performed simultaneously and/or in an order other than the order depicted in FIG. 5. In some embodiments, one or more operations are performed before, between, during, and/or after performing one or more operations of method **500**.

[0094] At operation **510**, in some embodiments, an IC layout diagram of a stored cell is obtained from a storage device, the stored cell including a functional circuit, e.g., a level shifter.

[0095] In some embodiments, obtaining the IC layout diagram of the stored cell from the storage device includes obtaining the IC layout diagram of the stored cell from cell library **607** of IC layout diagram generation system **600**, discussed below with respect to FIG. 6.

[0096] At operation **520**, a first portion of first through fourth pluralities of active areas and a plurality of gate regions of the cell are arranged as a functional circuit and a second portion of the first through fourth pluralities of active areas and plurality of gate regions are arranged as one of a decoupling capacitor or an antenna diode. In various embodiments, arranging the first portion as the functional circuit includes retaining an arrangement of the stored cell obtained from the storage device, modifying the arrangement of the stored cell, or performing an arrangement without obtaining the stored cell from the storage device.

[0097] Arranging the first portion of first through fourth pluralities of active areas and plurality of gate regions of the cell as the functional circuit and the second portion of the first through fourth pluralities of active areas and plurality of gate regions as the one of the decoupling capacitor or the antenna diode includes arranging the first portion of first through fourth pluralities of active regions/areas AA1-AA4 and plurality of gate regions G1-G4 and power rails VSS and VDD as first portion **200P1** and either the second portion of first through fourth pluralities of active regions/areas AA1-AA4 and plurality of gate regions G1-G4 and metal region M02 and via regions VG as second portion **200P2** as discussed above with respect to FIG. 2B or the second portion of first through fourth pluralities of active regions/areas AA1-AA4 and plurality of gate regions G1-G4 and metal region M03 and via regions VD as second portion **300P2** as discussed above with respect to FIG. 3B.

[0098] At operation **530**, in some embodiments, the IC layout diagram is stored in a storage device. In various embodiments, storing the IC layout diagram in the storage device includes storing the IC layout diagram in a non-volatile, computer-readable memory or a cell library, e.g., a database, and/or includes storing the IC layout diagram over a network. In some embodiments, storing the IC layout diagram in the storage device includes storing the IC layout diagram in cell library **607** or over network **614** of IC layout diagram generation system **600**, discussed below with respect to FIG. 6.

[0099] At operation **540**, in some embodiments, one or more manufacturing operations are performed based on the IC layout diagram. In some embodiments, performing one or more manufacturing operations includes performing one or more lithographic exposures based on the IC layout diagram. Performing one or more manufacturing operations, e.g., one or more lithographic exposures, based on the IC layout diagram is discussed above with respect to FIG. 4 and below with respect to FIG. 7.

[0100] By executing some or all of the operations of method **500**, an IC layout diagram is generated corresponding to an IC device in which a first portion includes features configured as a primary functional circuit, e.g., a level shifter, and a second portion includes features configured as a decoupling capacitor or antenna diode, thereby obtaining the benefits discussed above with respect to IC devices **100-300**.

[0101] FIG. 6 is a block diagram of IC layout diagram generation system **600**, in accordance with some embodiments. Methods described herein of designing IC layout diagrams in accordance with one or more embodiments are implementable, for example, using IC layout diagram generation

system **600**, in accordance with some embodiments.

[0102] In some embodiments, IC layout diagram generation system **600** is a general purpose computing device including a hardware processor **602** and a non-transitory, computer-readable storage medium **604**. Computer-readable storage medium **604**, amongst other things, is encoded with, i.e., stores, computer program code **606**, i.e., a set of executable instructions. Execution of instructions **606** by hardware processor **602** represents (at least in part) an EDA tool which implements a portion or all of a method such as method **500** of generating an IC layout diagram described above with respect to FIG. 5 (hereinafter, the noted processes and/or methods).

[0103] Processor **602** is electrically coupled to computer-readable storage medium **604** via a bus **608**. Processor **602** is also electrically coupled to an I/O interface **610** by bus **608**. A network interface **612** is also electrically connected to processor **602** via bus **608**. Network interface **612** is connected to a network **614**, so that processor **602** and computer-readable storage medium **604** are capable of connecting to external elements via network **614**. Processor **602** is configured to execute computer program code **606** encoded in computer-readable storage medium **604** in order to cause the IC layout diagram generation system **600** to be usable for performing a portion or all of the noted processes and/or methods. In one or more embodiments, processor **602** is a central processing unit (CPU), a multi-processor, a distributed processing system, an application specific integrated circuit (ASIC), and/or a suitable processing unit.

[0104] In one or more embodiments, computer-readable storage medium **604** is an electronic, magnetic, optical, electromagnetic, infrared, and/or a semiconductor system (or apparatus or device). For example, computer-readable storage medium **604** includes a semiconductor or solid-state memory, a magnetic tape, a removable computer diskette, a random access memory (RAM), a read-only memory (ROM), a rigid magnetic disk, and/or an optical disk. In one or more embodiments using optical disks, computer-readable storage medium **604** includes a compact disk-read only memory (CD-ROM), a compact disk-read/write (CD-R/W), and/or a digital video disc (DVD).

[0105] In one or more embodiments, computer-readable storage medium **604** stores computer program code **606** configured to cause the IC layout diagram generation system **600** (where such execution represents (at least in part) the EDA tool) to be usable for performing a portion or all of the noted processes and/or methods. In one or more embodiments, computer-readable storage medium **604** also stores information which facilitates performing a portion or all of the noted processes and/or methods. In one or more embodiments, computer-readable storage medium **604** stores cell library **607** of cells including such cells as disclosed herein, e.g., IC cells **110-130**, **200**, and **300** discussed above with respect to FIGS. 1-5.

[0106] IC layout diagram generation system **600** includes I/O interface **610**. I/O interface **610** is coupled to external circuitry. In one or more embodiments, I/O interface **610** includes a keyboard, keypad, mouse, trackball, trackpad, touchscreen, and/or cursor direction keys for communicating information and commands to processor **602**.

[0107] IC layout diagram generation system **600** also includes network interface **612** coupled to processor **602**. Network interface **612** allows IC layout diagram generation system **600** to communicate with network **614**, to which one or more other computer systems are connected. Network interface **612** includes wireless network interfaces such as BLUETOOTH, WIFI, WIMAX, GPRS, or WCDMA; or wired network interfaces such as ETHERNET, USB, or IEEE-1364. In one or more embodiments, a portion or all of noted processes and/or methods, is implemented in two or more IC layout diagram generation systems **600**.

[0108] IC layout diagram generation system **600** is configured to receive information through I/O interface **610**. The information received through I/O interface **610** includes one or more of instructions, data, design rules, libraries of standard cells, and/or other parameters for processing by processor **602**. The information is transferred to processor **602** via bus **608**. IC layout diagram generation system **600** is configured to receive information related to a UI through I/O interface

610. The information is stored in computer-readable storage medium **604** as user interface (UI) **642**.

[0109] In some embodiments, a portion or all of the noted processes and/or methods is implemented as a standalone software application for execution by a processor. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a software application that is a part of an additional software application. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a plug-in to a software application. In some embodiments, at least one of the noted processes and/or methods is implemented as a software application that is a portion of an EDA tool. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a software application that is used by IC layout diagram generation system **600**. In some embodiments, a layout diagram which includes standard cells is generated using a tool such as VIRTUOSO® available from CADENCE DESIGN SYSTEMS, Inc., or another suitable layout generating tool.

[0110] In some embodiments, the processes are realized as functions of a program stored in a non-transitory computer readable recording medium. Examples of a non-transitory computer readable recording medium include, but are not limited to, external/removable and/or internal/built-in storage or memory unit, e.g., one or more of an optical disk, such as a DVD, a magnetic disk, such as a hard disk, a semiconductor memory, such as a ROM, a RAM, a memory card, and the like.

[0111] FIG. 7 is a block diagram of IC manufacturing system **700**, and an IC manufacturing flow associated therewith, in accordance with some embodiments. In some embodiments, based on an IC layout diagram, at least one of (A) one or more semiconductor masks or (B) at least one component in a layer of a semiconductor integrated circuit is fabricated using IC manufacturing system **700**.

[0112] In FIG. 7, IC manufacturing system **700** includes entities, such as a design house **720**, a mask house **730**, and an IC manufacturer/fabricator (“fab”) **750**, that interact with one another in the design, development, and manufacturing cycles and/or services related to manufacturing an IC device **760**. The entities in IC manufacturing system **700** are connected by a communications network. In some embodiments, the communications network is a single network. In some embodiments, the communications network is a variety of different networks, such as an intranet and the Internet. The communications network includes wired and/or wireless communication channels. Each entity interacts with one or more of the other entities and provides services to and/or receives services from one or more of the other entities. In some embodiments, two or more of design house **720**, mask house **730**, and IC fab **750** is owned by a single larger company. In some embodiments, two or more of design house **720**, mask house **730**, and IC fab **750** coexist in a common facility and use common resources.

[0113] Design house (or design team) **720** generates an IC design layout diagram **722**. IC design layout diagram **722** includes various geometrical patterns, e.g., a IC cell **110-130**, **200** and/or **300** and/or IC layout **100** discussed above with respect to FIGS. 1-5. The geometrical patterns correspond to patterns of metal, oxide, or semiconductor layers that make up the various components of IC device **760** to be fabricated. The various layers combine to form various IC features. For example, a portion of IC design layout diagram **722** includes various IC features, such as an active region, gate electrode, source and drain, metal lines or vias of an interlayer interconnection, and openings for bonding pads, to be formed in a semiconductor substrate (such as a silicon wafer) and various material layers disposed on the semiconductor substrate. Design house **720** implements a proper design procedure to form the IC design layout diagram **722**. The design procedure includes one or more of logic design, physical design or place and route. IC design layout diagram **722** is presented in one or more data files having information of the geometrical patterns. For example, IC design layout diagram **722** can be expressed in a GDSII file format or DFII file format.

[0114] Mask house **730** includes mask data preparation **732** and mask fabrication **744**. Mask house

730 uses IC design layout diagram **722** to manufacture one or more masks **745** to be used for fabricating the various layers of IC device **760** according to IC design layout diagram **722**. Mask house **730** performs mask data preparation **732**, where IC design layout diagram **722** is translated into a representative data file (RDF). Mask data preparation **732** provides the RDF to mask fabrication **744**. Mask fabrication **744** includes a mask writer. A mask writer converts the RDF to an image on a substrate, such as a mask (reticle) **745** or a semiconductor wafer **753**. The IC design layout diagram **722** is manipulated by mask data preparation **732** to comply with particular characteristics of the mask writer and/or requirements of IC fab **750**. In FIG. 7, mask data preparation **732** and mask fabrication **744** are illustrated as separate elements. In some embodiments, mask data preparation **732** and mask fabrication **744** can be collectively referred to as mask data preparation.

[0115] In some embodiments, mask data preparation **732** includes optical proximity correction (OPC) which uses lithography enhancement techniques to compensate for image errors, such as those that can arise from diffraction, interference, other process effects and the like. OPC adjusts IC design layout diagram **722**. In some embodiments, mask data preparation **732** includes further resolution enhancement techniques (RET), such as off-axis illumination, sub-resolution assist features, phase-shifting masks, other suitable techniques, and the like or combinations thereof. In some embodiments, inverse lithography technology (ILT) is also used, which treats OPC as an inverse imaging problem.

[0116] In some embodiments, mask data preparation **732** includes a mask rule checker (MRC) that checks the IC design layout diagram **722** that has undergone processes in OPC with a set of mask creation rules which contain certain geometric and/or connectivity restrictions to ensure sufficient margins, to account for variability in semiconductor manufacturing processes, and the like. In some embodiments, the MRC modifies the IC design layout diagram **722** to compensate for limitations during mask fabrication **744**, which may undo part of the modifications performed by OPC in order to meet mask creation rules.

[0117] In some embodiments, mask data preparation **732** includes lithography process checking (LPC) that simulates processing that will be implemented by IC fab **750** to fabricate IC device **760**. LPC simulates this processing based on IC design layout diagram **722** to create a simulated manufactured device, such as IC device **760**. The processing parameters in LPC simulation can include parameters associated with various processes of the IC manufacturing cycle, parameters associated with tools used for manufacturing the IC, and/or other aspects of the manufacturing process. LPC takes into account various factors, such as aerial image contrast, depth of focus (“DOF”), mask error enhancement factor (“MEEF”), other suitable factors, and the like or combinations thereof. In some embodiments, after a simulated manufactured device has been created by LPC, if the simulated device is not close enough in shape to satisfy design rules, OPC and/or MRC are repeated to further refine IC design layout diagram **722**.

[0118] It should be understood that the above description of mask data preparation **732** has been simplified for the purposes of clarity. In some embodiments, mask data preparation **732** includes additional features such as a logic operation (LOP) to modify the IC design layout diagram **722** according to manufacturing rules. Additionally, the processes applied to IC design layout diagram **722** during mask data preparation **732** may be executed in a variety of different orders.

[0119] After mask data preparation **732** and during mask fabrication **744**, a mask **745** or a group of masks **745** are fabricated based on the modified IC design layout diagram **722**. In some embodiments, mask fabrication **744** includes performing one or more lithographic exposures based on IC design layout diagram **722**. In some embodiments, an electron-beam (e-beam) or a mechanism of multiple e-beams is used to form a pattern on a mask (photomask or reticle) **745** based on the modified IC design layout diagram **722**. Mask **745** can be formed in various technologies. In some embodiments, mask **745** is formed using binary technology. In some embodiments, a mask pattern includes opaque regions and transparent regions. A radiation beam,

such as an ultraviolet (UV) or EUV beam, used to expose the image sensitive material layer (e.g., photoresist) which has been coated on a wafer, is blocked by the opaque region and transmits through the transparent regions. In one example, a binary mask version of mask **745** includes a transparent substrate (e.g., fused quartz) and an opaque material (e.g., chromium) coated in the opaque regions of the binary mask. In another example, mask **745** is formed using a phase shift technology. In a phase shift mask (PSM) version of mask **745**, various features in the pattern formed on the phase shift mask are configured to have proper phase difference to enhance the resolution and imaging quality. In various examples, the phase shift mask can be attenuated PSM or alternating PSM. The mask(s) generated by mask fabrication **744** is used in a variety of processes. For example, such a mask(s) is used in an ion implantation process to form various doped regions in semiconductor wafer **753**, in an etching process to form various etching regions in semiconductor wafer **753**, and/or in other suitable processes.

[0120] IC fab **750** is an IC fabrication business that includes one or more manufacturing facilities for the fabrication of a variety of different IC products. In some embodiments, IC fab **750** is a semiconductor foundry. For example, there may be a manufacturing facility for the front end fabrication of a plurality of IC products (front-end-of-line (FEOL) fabrication), while a second manufacturing facility may provide the back end fabrication for the interconnection and packaging of the IC products (back-end-of-line (BEOL) fabrication), and a third manufacturing facility may provide other services for the foundry business.

[0121] IC fab **750** includes wafer fabrication tools **752** configured to execute various manufacturing operations on semiconductor wafer **753** such that IC device **760** is fabricated in accordance with the mask(s), e.g., mask **745**. In various embodiments, wafer fabrication tools **752** include one or more of a wafer stepper, an ion implanter, a photoresist coater, a process chamber, e.g., a CVD chamber or LPCVD furnace, a CMP system, a plasma etch system, a wafer cleaning system, or other manufacturing equipment capable of performing one or more suitable manufacturing processes as discussed herein.

[0122] IC fab **750** uses mask(s) **745** fabricated by mask house **730** to fabricate IC device **760**. Thus, IC fab **750** at least indirectly uses IC design layout diagram **722** to fabricate IC device **760**. In some embodiments, semiconductor wafer **753** is fabricated by IC fab **750** using mask(s) **745** to form the IC device **760**. In some embodiments, the IC fabrication includes performing one or more lithographic exposures based at least indirectly on IC design layout diagram **722**. Semiconductor wafer **753** includes a silicon substrate or other proper substrate having material layers formed thereon. Semiconductor wafer **753** further includes one or more of various doped regions, dielectric features, multilevel interconnects, and the like (formed at subsequent manufacturing steps).

[0123] Details regarding an IC manufacturing system (e.g., IC manufacturing system **700** of FIG. 7), and an IC manufacturing flow associated therewith are found, e.g., in U.S. Pat. No. 9,256,709, granted Feb. 9, 2016, U.S. Pre-Grant Publication No. 2015/0278429 A1, published Oct. 1, 2015, U.S. Pre-Grant Publication No. 2014/0040838 A1, published Feb. 6, 2014, and U.S. Pat. No. 7,260,442, granted Aug. 21, 2007, the entireties of each of which are hereby incorporated by reference.

[0124] In some embodiments, an integrated circuit (IC) device includes: a first gate structure and a second gate structure spaced apart in a first direction; a first conductor and a second conductor each extending in the first direction, spaced apart in a second direction, and overlying at least a portion of each of the first and second gate structures; a plurality of gate structures between the first and second gate structures; a conductive segment extending in the first direction between the first and second conductors and overlying the plurality of gate structures; a first plurality of active areas aligned along the first direction between the first and second gate structures; and a second plurality of active areas aligned along the first direction between the first and second gate structures. The first and second pluralities of active areas are between the first conductor and the second conductor. A first portion of the plurality of gate structures, a first portion of the first plurality of active areas,

and the second plurality of active areas are included in a functional circuit. A second portion of the plurality of gate structures, a second portion of the first plurality of active areas, and the conductive segment are included in a decoupling capacitor or an antenna diode.

[0125] In some embodiments, a method of fabricating an integrated circuit (IC) device includes: forming a first plurality of active areas aligned along a first direction; forming a second plurality of active areas aligned along the first direction and spaced apart from the first plurality of active areas in a second direction; forming a first gate structure, a second gate structure, and a plurality of gate structures between the first and second gate structures; and forming a first conductor, a second conductor, and a conductive segment each extending in the first direction. The first gate structure is formed at first ends of the first and second pluralities of active areas. The second gate structure is formed at second ends of the first and second pluralities of active areas. The first and second conductors are formed to overlie at least a portion of each of the first and second gate structures. The first and second conductors are formed such that the first and second pluralities of active areas are between the first conductor and the second conductor. The conductive segment is formed between the first and second conductors and overlying the plurality of gate structures. A functional circuit is configured to include a first portion of the plurality of gate structures, a first portion of the first plurality of active areas, and the second plurality of active areas. A decoupling capacitor or an antenna diode is configured to include a second portion of the plurality of gate structures, a second portion of the first plurality of active areas, and the conductive segment.

[0126] In some embodiments, an integrated circuit (IC) device includes: a functional circuit that includes a first portion of a plurality of gate structures, a first portion of a first plurality of active areas, and a second plurality of active areas; at least one of a decoupling capacitor or an antenna diode that includes a second portion of the plurality of gate structures, a second portion of the first plurality of active areas, and a conductive segment; a first gate structure and a second gate structure spaced apart in a first direction; and a first conductor and a second conductor each extending in the first direction, spaced apart in a second direction, and overlying at least a portion of each of the first and second gate structures. The first and second portions of the plurality of gate structures are between the first and second gate structures. The conductive segment extends over the second portion of the plurality of gate structures between the first and second conductors. The first and second pluralities of active areas extend along the first direction between the first and second gate structures.

[0127] It will be readily seen by one of ordinary skill in the art that one or more of the disclosed embodiments fulfill one or more of the advantages set forth above. After reading the foregoing specification, one of ordinary skill will be able to affect various changes, substitutions of equivalents and various other embodiments as broadly disclosed herein. It is therefore intended that the protection granted hereon be limited only by the definition contained in the appended claims and equivalents thereof.

Claims

1. An integrated circuit (IC) device comprising: a first gate structure and a second gate structure spaced apart in a first direction; a first conductor and a second conductor each extending in the first direction, spaced apart in a second direction, and overlying at least a portion of each of the first and second gate structures; a plurality of gate structures between the first and second gate structures; a conductive segment extending in the first direction between the first and second conductors and overlying the plurality of gate structures; a first plurality of active areas aligned along the first direction between the first and second gate structures; and a second plurality of active areas aligned along the first direction between the first and second gate structures, wherein: the first and second pluralities of active areas are between the first conductor and the second conductor, a first portion of the plurality of gate structures, a first portion of the first plurality of active areas, and the second

plurality of active areas are included in a functional circuit, and a second portion of the plurality of gate structures, a second portion of the first plurality of active areas, and the conductive segment are included in a decoupling capacitor or an antenna diode.

2. The IC device of claim 1, wherein: the first portion of the plurality of gate structures, the first portion of the first plurality of active areas, and the second plurality of active areas include: a first subset of the active areas of the first plurality of active areas; and an entirety of the second plurality of active areas, and the second portion of the plurality of gate structures, the second portion of the first plurality of active areas, and the conductive segment include: a second subset of the active areas of the first plurality of active areas.

3. The IC device of claim 2, wherein: the first subset of the active areas of the first plurality of active areas includes: first and second active areas, and the second subset of the active areas of the first plurality of active areas includes: a third active area between the first and second active areas.

4. The IC device of claim 2, further comprising: first, second, and third n-wells, wherein: the second plurality of active areas includes at least one active area in each of the first, second, and third n-wells.

5. The IC device of claim 1, wherein: the first portion of the plurality of gate structures, the first portion of the first plurality of active areas, and the second plurality of active areas include: a first plurality of gate structure portions, and the second portion of the plurality of gate structures, the second portion of the first plurality of active areas, and the conductive segment include: a second plurality of gate structure portions aligned in the second direction with, and electrically isolated from, the first plurality of gate structure portions.

6. The IC device of claim 1, wherein: the conductive segment is configured to carry a power supply voltage, and the second portion of the plurality of gate structures, the second portion of the first plurality of active areas, and the conductive segment include: a plurality of via structures configured to couple the conductive segment to corresponding gate structures of the second portion of the plurality of gate structures.

7. The IC device of claim 1, wherein: the conductive segment is configured to carry an input signal of the functional circuit, and the second portion of the plurality of gate structures, the second portion of the first plurality of active areas, and the conductive segment include: a plurality of via structures configured to couple the conductive segment to corresponding source/drain structures in one or more active areas of the first plurality of active areas.

8. The IC device of claim 1, wherein the functional circuit includes a level shifter.

9. A method of fabricating an integrated circuit (IC) device, the method comprising: forming a first plurality of active areas aligned along a first direction; forming a second plurality of active areas aligned along the first direction and spaced apart from the first plurality of active areas in a second direction; forming a first gate structure, a second gate structure, and a plurality of gate structures between the first and second gate structures; and forming a first conductor, a second conductor, and a conductive segment each extending in the first direction, wherein: the first gate structure is formed at first ends of the first and second pluralities of active areas, the second gate structure is formed at second ends of the first and second pluralities of active areas, the first and second conductors are formed to overlie at least a portion of each of the first and second gate structures, the first and second conductors are formed such that the first and second pluralities of active areas are between the first conductor and the second conductor, the conductive segment is formed between the first and second conductors and overlying the plurality of gate structures, a functional circuit is configured to include a first portion of the plurality of gate structures, a first portion of the first plurality of active areas, and the second plurality of active areas, and a decoupling capacitor or an antenna diode is configured to include a second portion of the plurality of gate structures, a second portion of the first plurality of active areas, and the conductive segment.

10. The method of claim 9, wherein: the forming the first plurality of active areas and the forming the second plurality of active areas include: arranging a first subset of the first plurality of active

areas and an entirety of the second plurality of active areas in the functional circuit, and the forming the first plurality of active areas further includes: arranging a second subset of the first plurality of active areas in one of the decoupling capacitor or the antenna diode.

11. The method of claim 9, wherein: the forming the second plurality of active areas includes: positioning active areas of the second plurality of active areas in each of first, second, and third n-wells.

12. The method of claim 9, wherein the forming the plurality of gate structures includes: forming a first plurality of gate structure portions in the functional circuit; and forming a second plurality of gate structure portions in the decoupling capacitor or the antenna diode, the second plurality of gate structure portions being formed to be aligned in the second direction with, and electrically isolated from, the first plurality of gate structure portions.

13. An integrated circuit (IC) device comprising: a functional circuit that includes a first portion of a plurality of gate structures, a first portion of a first plurality of active areas, and a second plurality of active areas; at least one of a decoupling capacitor or an antenna diode that includes a second portion of the plurality of gate structures, a second portion of the first plurality of active areas, and a conductive segment; a first gate structure and a second gate structure spaced apart in a first direction; and a first conductor and a second conductor each extending in the first direction, spaced apart in a second direction, and overlying at least a portion of each of the first and second gate structures, wherein: the first and second portions of the plurality of gate structures are between the first and second gate structures, the conductive segment extends over the second portion of the plurality of gate structures between the first and second conductors, and the first and second pluralities of active areas extend along the first direction between the first and second gate structures.

14. The IC device of claim 13, wherein: the first portion of the first plurality of active areas includes: a first active area and a second active area, which are spaced apart in the first direction, and the second portion of the first plurality of active areas includes: a third active area between the first and second active areas.

15. The IC device of claim 13, further comprising: first, second, and third n-wells, wherein: the second plurality of active areas includes at least one active area in each of the first, second, and third n-wells.

16. The IC device of claim 15, wherein: the second n-well is between the first and third n-wells, the second n-well has a first length in the first direction, and the conductive segment has a second length in the first direction that is greater than the first length.

17. The IC device of claim 13, wherein: the first portion of the plurality of gate structures is aligned in the second direction with, and electrically isolated from, the second portion of the plurality of gate structures.

18. The IC device of claim 13, wherein: the conductive segment is configured to carry a power supply voltage, and the IC device further comprises a plurality of via structures coupling the conductive segment to corresponding gate structures of the second portion of the plurality of gate structures.

19. The IC device of claim 13, wherein: the conductive segment is configured to carry an input signal of the functional circuit, and the IC device further comprises a plurality of via structures coupling the conductive segment to corresponding source/drain structures in one or more active areas of the second portion of the first plurality of active areas.

20. The IC device of claim 13, wherein the functional circuit includes a level shifter.
